Mississippi Valley State University

# CS 341 — Discrete Structures

Practice Set 41: Approved Solutions

*Rooted and Spanning Trees*

Covers Chartrand & Zhang, Section 13.2

**Status of this set**

**Ungraded practice.** Solutions released with it. **The last practice set of the course.**

# Part I: How to Practice This Material

**The three facts**

1. A rooted tree's vocabulary is all defined by **distance from the root**.
2. A connected graph of order $n$ and size $m$ needs $m - n + 1$ edges removed to leave a spanning tree.
3. A rooted tree where every vertex has at most $c$ children and height $h$ has order at most $1 + c + c^2 + \cdots + c^h$.

**Three mistakes to expect**

1. **Confusing height with order.** Height counts levels below the root.
2. **Off-by-one on levels.** The root is level 0, so height $h$ means $h + 1$ levels.
3. **Computing $m - n$ instead of $m - n + 1$.**

# Part II: Practice Problems

## Section A — Rooted trees

**1.** A tree has root $v_1$ and edges $v_1v_2$, $v_1v_3$, $v_2v_4$, $v_3v_5$, $v_3v_6$, $v_5v_7$.

(a) List the vertices at each level, and give the height.

(b) Give the parent of $v_7$ and the children of $v_3$.

(c) List the leaves and the ancestors of $v_7$.

**Approved solution**

**(a)** Level 0: $v_1$. Level 1: $v_2, v_3$. Level 2: $v_4, v_5, v_6$. Level 3: $v_7$. **Height 3.**
**(b)** Parent of $v_7$ is $v_5$. Children of $v_3$ are $v_5$ and $v_6$.
**(c)** Leaves: $v_4, v_6, v_7$. Ancestors of $v_7$: $v_5, v_3, v_1$ — every vertex on the $v_1$–$v_7$ path other than $v_7$ itself.

Order 7, size $6 = n - 1$. ✓ Note $v_2$ has degree 2 and is neither a leaf nor the root — it is an interior vertex with one child.

**2.**

(a) What is the maximum order of a binary tree of height 5?

(b) A rooted tree has every vertex with at most 3 children and height 4. What is its maximum order?

(c) A binary tree has 1000 vertices. What is the least possible height?

**Approved solution**

**(a)** $1 + 2 + 4 + 8 + 16 + 32 = 63 = 2^6 - 1$.
**(b)** $1 + 3 + 9 + 27 + 81 = 121$.
**(c)** We need $2^{h+1} - 1 \geq 1000$, so $2^{h+1} \geq 1001$. Since $2^9 = 512$ and $2^{10} = 1024$, we need $h + 1 \geq 10$, so $h \geq 9$. **The least height is 9.**

**Part (c) is why binary search is logarithmic.** A tree holding $n$ items cannot be shorter than about $\log_2 n$, and the height is the number of comparisons.

## Section B — Spanning trees

**3.** For each connected graph, give the size of a spanning tree and the number of edges that must be removed.

(a) Order 15, size 28.

(b) $K_7$.

(c) A 5-regular graph of order 12.

**Approved solution**

**(a)** Spanning tree has 14 edges; remove $28 - 14 = 14$.
**(b)** $K_7$ has order 7 and size $\binom{7}{2} = 21$. Spanning tree has 6 edges; remove $21 - 6 = 15$.
**(c)** Size is $\frac{12 \cdot 5}{2} = 30$. Spanning tree has 11 edges; remove $30 - 11 = 19$.

In (c) the size came from the First Theorem of Graph Theory: the degree sum 60 equals $2m$.

**4.** Let $G$ be the graph on $V = \{1, \ldots, 5\}$ with $E = \{12, 13, 23, 34, 35, 45\}$.

(a) How many edges must be removed to leave a spanning tree?

(b) Give one spanning tree by listing its edges.

(c) Does $G$ have more than one spanning tree? Explain without listing them all.

**Approved solution**

**(a)** Order 5, size 6. A spanning tree has 4 edges, so remove 2.
**(b)** For instance $\{12, 13, 34, 45\}$ — connected, 4 edges, no cycle. (Many others work.)
**(c) Yes.** $G$ contains the cycle $(1, 2, 3, 1)$, and deleting any one of its three edges leaves a connected graph. Different choices give different spanning trees, so there are at least three.

**In general a graph has one spanning tree exactly when it is already a tree.** Any extra edge lies on a cycle, and that cycle offers a choice.

## Section C — Writing

**5.** In four to six complete sentences, explain why a graph has a spanning tree if and only if it is connected, and describe a procedure for finding one.

**Approved solution**

**Model answer.** If a graph has a spanning tree, then that tree uses every vertex and is connected, so any two vertices are joined by a path within the tree and therefore within the graph — so the graph is connected. Conversely, suppose the graph is connected. While any cycle remains, delete one of its edges; deleting an edge that lies on a cycle cannot disconnect the graph, because the rest of the cycle still joins its endpoints. The process must terminate, since each step removes an edge and there are finitely many. What remains is connected, has no cycles, and still contains every vertex, so it is a spanning tree.

**Full credit requires** both directions and the reason deleting a cycle edge is safe. The termination remark is a bonus.

**Instructor notes**

- **Problem 2(c)** is the payoff of the section: the $\log_2 n$ height bound is why binary search costs what it does. Worth stating explicitly on the last teaching day.
- **Problem 4(c)** gives the clean characterisation — a unique spanning tree means the graph was already a tree.
- **Problem 5** contains the standard construction and the reason it is safe.
- **This is the final practice set.** Direct students to the review sheet and the practice final next.